DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE, RAIGAD
Supplementary Summer 2025

Branch Name: Common to All Branches   Class: FY

Subject Name: Engineering Graphics   Subject Code: BTES203G

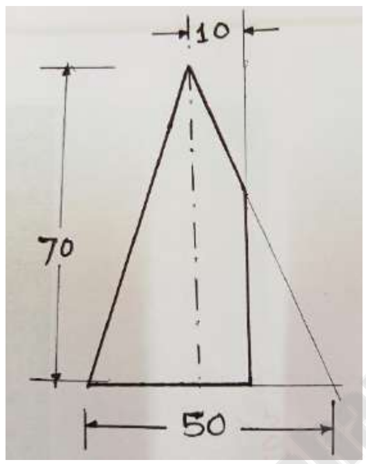
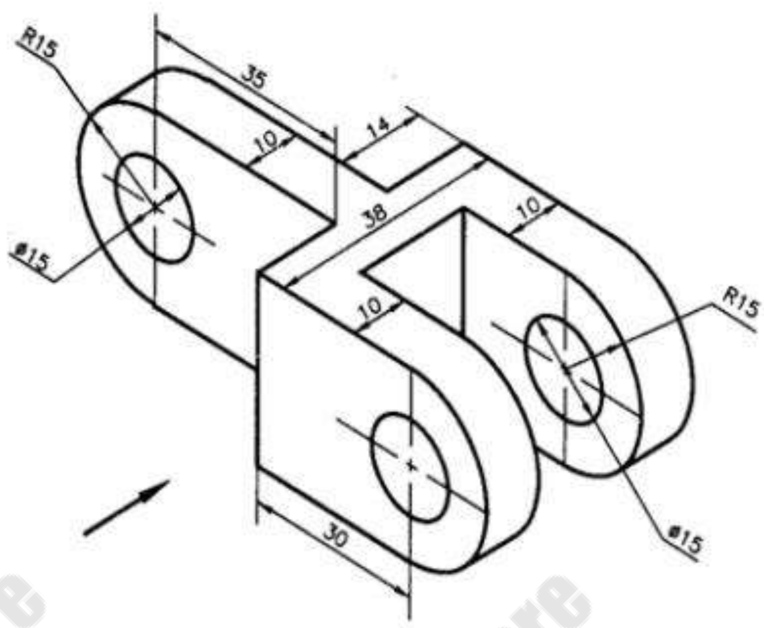
Max Marks: 60   Duration: 04Hrs   Date:

Instructions:-

1. ***Question No. 1** is compulsory and include objective-type questions.*
2. *Attempt **Any FOUR** questions out of Q.No 02 to Q. No 6.*
3. *Assume the suitable data wherever necessary.*
4. *Mention the Question numbers correctly on the drawing sheets.*

Q.1	**Objective type questions. (Compulsory Question)**	Level	**Marks**
1	True length of Profile Line can be observed in….. a) Front View b) Top view c) Side View d) None of the above	CO1	1
2.	The included angle of Regular Hexagonal plane is a)118^0 b) 60^0 c) 102^0 d) 120^0	CO2	1
3	Frontal Line in shows Point View in…….. a) Front View b) Top view c) Side View d) May be Front or Side view	CO1	1
4.	Point 'K' has both Front view & Top view Below XY line, Point 'K' is located in a) 3^{rd} Quadrant b) 4^{th} Quadrant c) 2^{nd} Quadrant d) 1^{st} Quadrant	CO1	1
5.	Cone resting on its base on ground shows……… shape in Top View. a) Ellipse b) Square c) Rectangular d) circular	CO3	1
6.	The pyramids are having ………shape for its slant faces. a) Triangular b) Square c) Rectangular d) circular	CO3	1
7.	Following is the plane with 04 sides …. a) Heptagon b) Octagon c) Trapezium d) Scalene	CO2	1
8.	The plane parallel to FRP shows … a) True shape in FV b) Edge view in Side view c) Edge view in TV d) All above	CO2	1

9	The line used for showing the details after cutting of the object is … a) Section line b) Dotted line c) center line d) Thin line	CO4	1
10	Quarter Circle in Isometric Drawing can be drawn …. a) Using 2 centers b) Using Only one center c) Using 4 centers d) None of the Above	CO5	1
11	Cylinder is kept on ground on its base. As per Third Angle method it shows….. a) Circle below XY b) Circle above XY c) Rectangle below XY d) Both (b) & (c)	CO5	1
12	The LHSV according to First Angle Method of projection is drawn at…. a) Right side of FV b) Right side of TV c) Left side of FV d) Left side of TV	CO5	1
Q.2	**Attempt the following questions (Any02)**		
a.	Draw the Regular Heptagon of 50mm side by any method.	CO-1	06
b.	Using Drawing Standards, DRAW the following statement: **Engineers Create The World**	CO-1	06
c.	What are the different types of Methods of Dimensions used in Drawing. Draw the various types of lines used in Engineering drawing and state its use,	CO1	06
Q.3	**Attempt the following questions (Any 02)**		
a.	Horizontal line AB 70mm long and Perpendicular to VP. Point A is 20mm above HP and 15mm in front of VP. Complete the projections and find its Plan Length and Elevation Length and locate the traces.	CO2	06
b.	The Elevation length of line PQ is 55mm and makes an angle of 50^0 with reference plane. Its True length measures 80mm and line is inclined at 40^0 with VP. Point P is at 15mm from both the reference planes. Complete the projections and locate the traces.	CO2	06
c.	Locate the positions of the following points: i)Point A in HP and 30mm in front of VP ii)Point B 10 mm below HP and 20mm behind VP iii)Point C in VP and 25 mm above HP	CO2	06
Q.4	**Attempt the following question (Any 02)**		
a.	A regular Pentagon of 50mm side is resting in HP on one of its edges with surface inclined to HP by 45^0. Complete the projections if resting edge makes an angle of 35^0 with VP	CO3	06
b	A Circular Plate resting on point of circumference in VP and inclined to VP by 45^0.Complete the projections if front view of the diagonal passing through resting point makes an angle of 40^0 with HP.	CO3	06
c.	A Square plate of size 50mm is resting on one of its corners in HP with diagonal inclined at 40^0 with HP and 30^0 with VP. Complete the projections.	CO3	06

Q.5	**Attempt the following question (Any 02)**		
a.	A Cylinder of diameter 50 mm and height of Axis 75 mm is resting on base point in HP with the axis inclined to HP by 45^0 and inclined to VP by 30^0. Complete the projections if Top face is towards the observer.	CO4	06
b.	A Hexagonal Pyramid of base side 40mm and height of axis 85 mm, is resting on base in HP with one of the base edges parallel to VP . It is cut by an AIP 30^0 which cuts the axis at point 25mm from the apex. Draw the FV, TV and True shape of the section.	CO4	06
c.	Develop the lateral surface of retained large part of cone as shown in Fig-1.  Fig-1	CO4	06
Q.6	**Attempt the following question (Any 02)**		
a.	Draw the Front view in Arrow direction, Top view and Side view of the following sketch by **First angle** Method of Projection **(Fig.2)**  **Fig:2** All dimensions are in mm	CO5	6

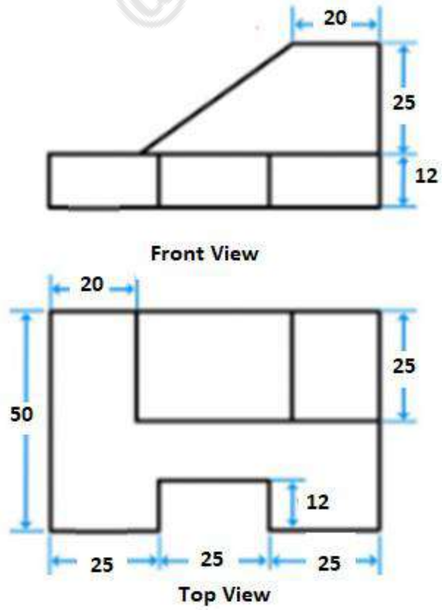
<table>
<tr><td>b.</td><td>Draw the Isometric View of the following object (Fig-3).</td><td></td><td></td></tr>
<tr><td></td><td>

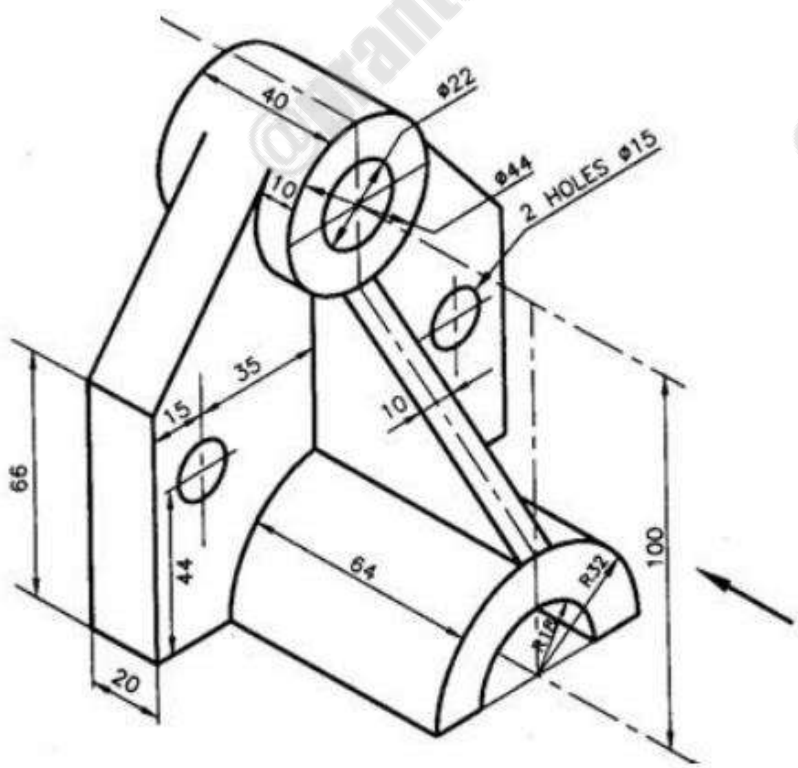
Fig-3</td><td>CO5</td><td>6</td></tr>
<tr><td>c.</td><td>Draw the Front view in Arrow- direction, Top view and Side view of the following sketch by Third angle Method of Projection (Fig.4)


Fig:4 All dimensions are in mm</td><td>CO5</td><td>6</td></tr>
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